

# 1. INTRODUCTION

## 1.1 Project Overview

The **Automated Employee Onboarding & Offboarding System** is a digital solution designed to streamline and automate the processes involved when employees join or leave an organization. Traditional onboarding and offboarding activities often require significant manual effort, leading to delays, errors, and compliance risks. This system automates these tasks through predefined workflows, ensuring that all required activities are completed efficiently and consistently.

The system manages key processes such as employee record creation, account provisioning, equipment allocation, document verification, approval workflows, access management, asset retrieval, and account deactivation. By integrating these activities into a single platform, the system improves operational efficiency, enhances security, and provides a better experience for employees and administrators.

## 1.2 Purpose

The purpose of the **Automated Employee Onboarding & Offboarding System** is to automate and standardize employee lifecycle management from hiring to exit. The project aims to reduce manual work, minimize processing time, eliminate errors, and ensure organizational policies and compliance requirements are consistently followed.

The main objectives of this project are:

- Automate employee onboarding and offboarding workflows.
- Reduce manual paperwork and administrative effort.
- Ensure timely approvals and task completion.
- Improve data accuracy and process transparency.
- Enhance security by automatically granting and revoking system access.
- Track employee requests and workflow progress in real time.
- Improve the overall employee experience through faster and more efficient processes.

## 2. IDEATION PHASE

### 2.1 Problem Statement

Organizations often manage employee onboarding and offboarding through manual processes involving paperwork, emails, and multiple departments. These traditional methods are time-consuming, prone to human error, and can result in delayed approvals, incomplete documentation, security risks, and poor employee experiences. During onboarding, delays in creating user accounts, assigning assets, and granting system access can reduce employee productivity. Similarly, during offboarding, failure to deactivate accounts or recover company assets on time can create security and compliance issues.

The **Automated Employee Onboarding & Offboarding System** addresses these challenges by automating workflow processes, approval management, access provisioning, asset tracking, and notifications. This ensures a faster, more accurate, secure, and standardized employee lifecycle management process.

### 2.2 Empathy Map Canvas

#### User: HR Manager

##### Says

- I want onboarding and offboarding to be quick and error-free.
- I need all approvals completed on time.
- I want to track employee progress easily.

##### Thinks

- Manual processes take too much time.
- Missing approvals can create compliance issues.
- Employee information should always be accurate.

##### Does

- Creates employee records.
- Sends approval requests.

- Assigns tasks to IT and other departments.
- Monitors workflow status.

### **Feels**

- Frustrated by repetitive manual tasks.
- Concerned about security and compliance.
- Satisfied when processes are completed smoothly and on time.

## **User: Employee**

### **Says**

- I want my accounts and resources ready on my first day.
- I expect a smooth joining and exit process.

### **Thinks**

- Delays affect my productivity.
- A simple process creates a better work experience.

### **Does**

- Completes required forms and documents.
- Tracks onboarding or offboarding progress.
- Responds to notifications and approvals.

### **Feels**

- Excited when joining the organization.
- Confident when the process is transparent.
- Appreciated when tasks are completed efficiently.

## **2.3 Brainstorming**

During the brainstorming phase, several ideas were evaluated to improve employee lifecycle management. The following solutions were identified as the most effective:

- Automate onboarding and offboarding workflows.
- Implement role-based approval processes.
- Automatically create and deactivate user accounts.
- Manage employee documents digitally.
- Track company assets assigned to employees.
- Send automatic email and system notifications.
- Provide dashboards for HR and managers to monitor progress.
- Generate reports for compliance and auditing.
- Maintain complete records of employee activities.
- Integrate with HR, IT, and Security departments for seamless coordination.

These ideas were combined to develop an integrated system that improves efficiency, reduces manual effort, enhances security, and provides a better experience for employees and administrators.

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map

| Stage               | Employee Actions                   | HR Actions                                | System Response                                |
|---------------------|------------------------------------|-------------------------------------------|------------------------------------------------|
| Job Offer           | Accepts offer                      | Creates employee profile                  | Employee record is generated                   |
| Pre-Onboarding      | Uploads required documents         | Reviews submitted documents               | Sends notifications and approval requests      |
| Onboarding          | Completes joining formalities      | Assigns tasks to IT and other departments | Creates user accounts and assigns assets       |
| Employment          | Uses company systems and resources | Monitors employee information             | Maintains employee records and workflow status |
| Offboarding Request | Submits resignation                | Initiates offboarding workflow            | Generates exit checklist and approval tasks    |
| Exit Process        | Returns company assets             | Verifies asset collection and approvals   | Deactivates accounts and updates records       |

|            |                       |                         |                                        |
|------------|-----------------------|-------------------------|----------------------------------------|
| Completion | Receives confirmation | Closes employee profile | Archives records and generates reports |
|------------|-----------------------|-------------------------|----------------------------------------|

## 3.2 Solution Requirement

### Functional Requirements

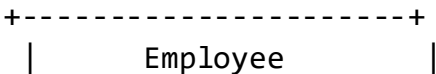
- Employee registration and profile management.
- Automated onboarding and offboarding workflows.
- Role-based approval process.
- User account creation and deactivation.
- Asset allocation and return tracking.
- Document upload and verification.
- Email and system notifications.
- Dashboard for HR, Managers, and IT teams.
- Report generation and audit tracking.
- Secure authentication and role-based access control.

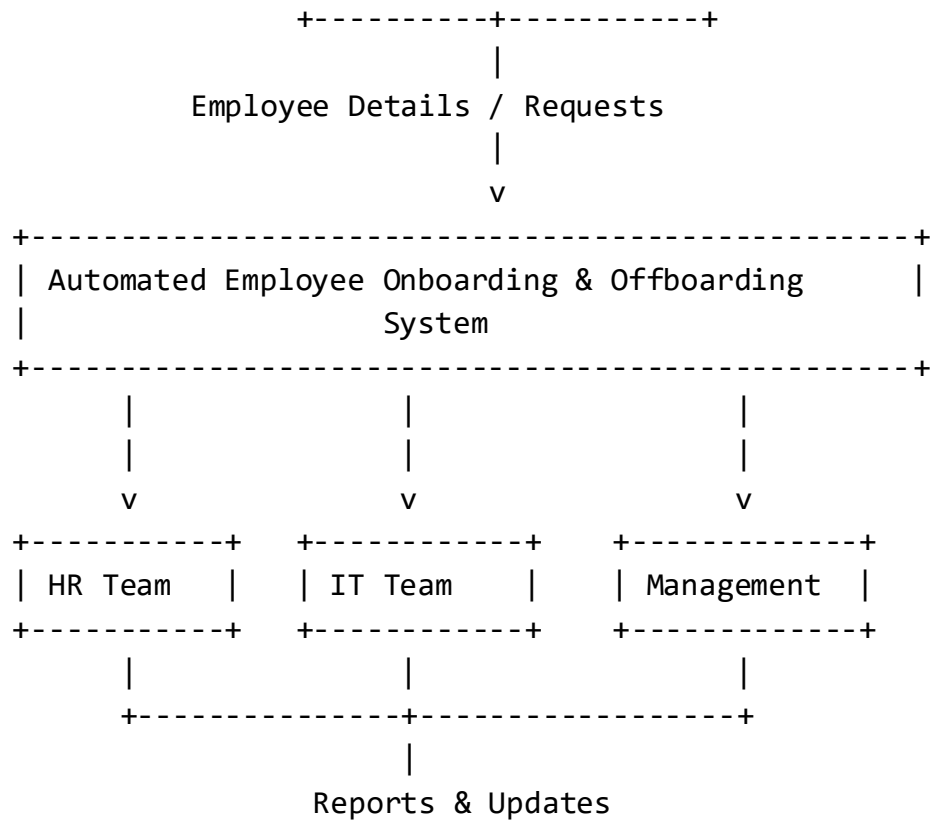
### Non-Functional Requirements

- High system availability and reliability.
- Fast workflow processing.
- Secure data storage and access.
- User-friendly interface.
- Scalability to support organizational growth.
- Data privacy and compliance.
- Easy maintenance and future enhancements.

## 3.3 Data Flow Diagram (DFD)

### Level 0 (Context Diagram)





## Level 1 DFD

1. Employee submits onboarding or offboarding request.
2. HR verifies employee information.
3. System sends approval requests.
4. IT provisions or revokes system access.
5. Assets are assigned or collected.
6. Notifications are sent to stakeholders.
7. Employee records are updated.
8. Reports are generated for auditing.

## 3.4 Technology Stack

| Layer    | Technology |
|----------|------------|
| Platform | ServiceNow |

|                      |                                                |
|----------------------|------------------------------------------------|
| Development          | ServiceNow App Engine Studio                   |
| Workflow Automation  | Flow Designer                                  |
| Programming Language | JavaScript (Server-side scripting)             |
| Database             | ServiceNow Tables                              |
| User Interface       | Service Portal / Workspace                     |
| Authentication       | Role-Based Access Control (RBAC)               |
| Notifications        | Email Notifications                            |
| Reporting            | ServiceNow Reports & Dashboards                |
| Version Control      | ServiceNow Update Sets                         |
| Browser Support      | Google Chrome, Microsoft Edge, Mozilla Firefox |
| Operating System     | Windows 10/11                                  |

The technology stack provides a secure, scalable, and efficient platform for automating employee onboarding and offboarding processes. ServiceNow's built-in workflow automation, database management, reporting capabilities, and role-based security help streamline HR operations while ensuring compliance and improving the overall employee experience.

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The **Automated Employee Onboarding & Offboarding System** is designed to address the limitations of traditional manual HR processes. Manual onboarding and offboarding often result in delays, paperwork, communication gaps, security risks, and inconsistent execution of tasks. The proposed solution automates the complete employee lifecycle by integrating HR, IT, and management activities into a single workflow.

| Problem           | Proposed Solution                       | Benefit              |
|-------------------|-----------------------------------------|----------------------|
| Manual paperwork  | Digital forms and automated workflows   | Faster processing    |
| Delayed approvals | Workflow-based approval system          | Reduced waiting time |
| Human errors      | Automated task execution and validation | Improved accuracy    |

|                         |                                                 |                         |
|-------------------------|-------------------------------------------------|-------------------------|
| Poor tracking           | Real-time dashboards and status monitoring      | Better visibility       |
| Security risks          | Automatic account provisioning and deactivation | Enhanced security       |
| Asset management issues | Asset allocation and return tracking            | Improved accountability |
| Compliance challenges   | Audit logs and automated reports                | Better compliance       |

The proposed system ensures a secure, efficient, and standardized employee onboarding and offboarding process while improving productivity and employee satisfaction.

## 4.2 Proposed Solution

The proposed solution is a **ServiceNow-based Automated Employee Onboarding & Offboarding System** that digitizes and automates employee lifecycle management. The system uses predefined workflows, role-based access control, automated notifications, and approval mechanisms to reduce manual effort and improve operational efficiency.

### Key Features

- Employee onboarding request management
- Employee offboarding request management
- Automated approval workflows
- User account creation and deactivation
- Document upload and verification
- Asset assignment and recovery tracking
- Email and system notifications
- Role-based access control (RBAC)
- Dashboard and reporting
- Audit trail for compliance and monitoring

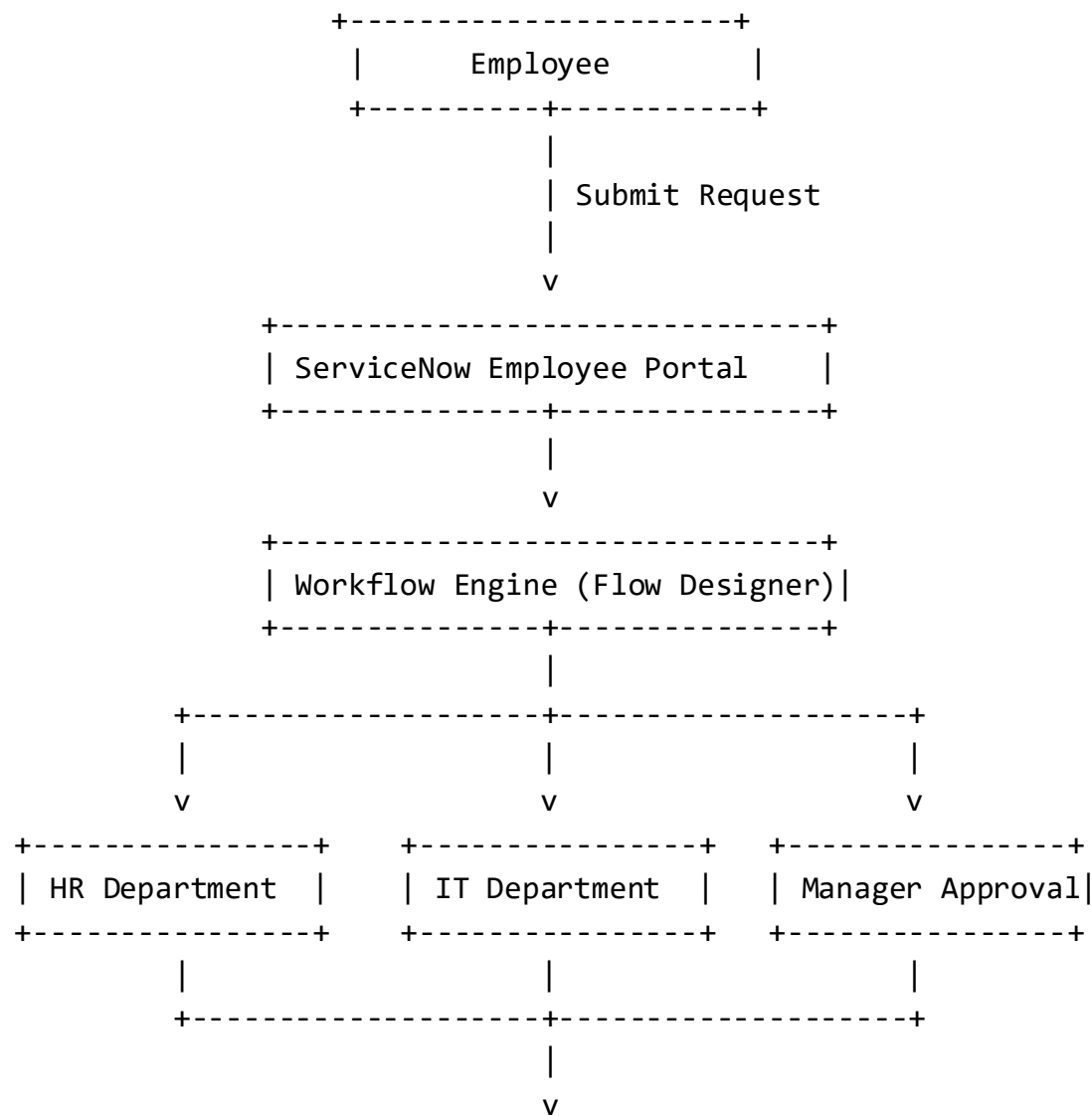
### Benefits

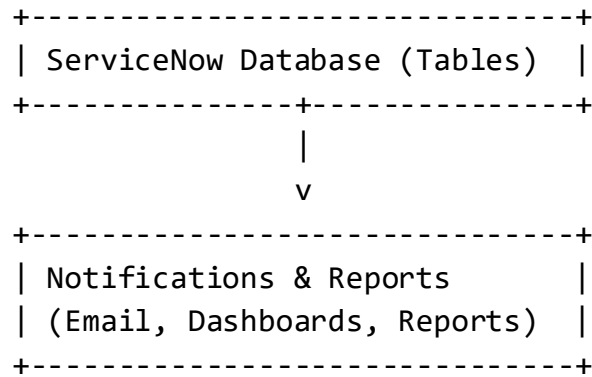
- Reduces manual work

- Improves workflow efficiency
- Minimizes processing errors
- Strengthens organizational security
- Enhances employee experience
- Provides real-time tracking and reporting
- Ensures compliance with organizational policies

## 4.3 Solution Architecture

### System Architecture





## Architecture Description

1. Employees submit onboarding or offboarding requests through the **ServiceNow Employee Portal**.
2. The **Flow Designer** automatically initiates the required workflow.
3. Approval requests are routed to the HR department, reporting manager, and IT team.
4. HR validates employee information and coordinates the process.
5. IT provisions or revokes system access and manages company assets.
6. All employee records are securely stored in **ServiceNow Tables**.
7. The system sends automated email notifications and generates dashboards and reports for monitoring and compliance.

This architecture provides a centralized, secure, and scalable solution that automates employee onboarding and offboarding while improving efficiency, transparency, and data security across the organization.

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

Project planning is an essential phase that defines the activities, resources, timeline, and milestones required for the successful development of the **Automated Employee Onboarding & Offboarding System**. A structured plan ensures that the project is completed within the specified schedule while meeting all functional and quality requirements.

## Project Objectives

- Develop an automated employee onboarding and offboarding system using the ServiceNow platform.
- Streamline HR processes through workflow automation.
- Reduce manual effort and processing time.
- Improve data accuracy and security.
- Enhance the overall employee experience.

## Project Phases

| Phase                        | Activities                                                                         | Duration |
|------------------------------|------------------------------------------------------------------------------------|----------|
| Requirement Gathering        | Analyze business requirements, identify stakeholders, define project scope         | 1 Week   |
| System Design                | Design workflows, database structure, user interface, and architecture             | 1 Week   |
| Development                  | Create ServiceNow application, forms, workflows, business rules, and notifications | 2 Weeks  |
| Testing                      | Perform functional, integration, and user acceptance testing (UAT)                 | 1 Week   |
| Deployment                   | Deploy the application and configure production settings                           | 1 Week   |
| Documentation & Presentation | Prepare project documentation, user manual, and presentation                       | 1 Week   |

## Resource Planning

| Resource            | Responsibility                                         |
|---------------------|--------------------------------------------------------|
| Project Team        | Application development and testing                    |
| HR Department       | Requirement validation and user acceptance testing     |
| IT Team             | System configuration and technical support             |
| ServiceNow Platform | Workflow automation, database, and application hosting |
| Project Guide       | Project review and guidance                            |

## Risk Management

| Risk                      | Mitigation Strategy                                           |
|---------------------------|---------------------------------------------------------------|
| Requirement changes       | Conduct regular review meetings with stakeholders             |
| Development delays        | Follow a planned project schedule and monitor progress        |
| System integration issues | Perform continuous integration and testing                    |
| Data security risks       | Implement role-based access control and secure authentication |
| User adoption challenges  | Provide training and user documentation                       |

## Project Deliverables

- ServiceNow application
- Automated onboarding workflow
- Automated offboarding workflow
- Employee and HR dashboards
- Reports and notifications
- Testing report
- User documentation
- Final project report and presentation

Proper planning and scheduling ensure that the project is completed efficiently, within the planned timeline, and in accordance with the required quality standards.

## 6. FUNCTIONAL AND PERFORMANCE TESTING

### 6.1 Performance Testing

Performance testing was conducted to evaluate the efficiency, reliability, and responsiveness of the **Automated Employee Onboarding & Offboarding System** under normal operating conditions. The objective was to ensure that the application performs consistently while handling employee onboarding and offboarding requests without delays or failures.

## Objectives of Performance Testing

- Measure system response time.
- Verify workflow execution speed.
- Evaluate system stability during multiple user requests.
- Ensure efficient database operations.
- Confirm reliable notification delivery.
- Validate overall system performance.

## Performance Test Scenarios

| Test Scenario              | Expected Result                                       | Status |
|----------------------------|-------------------------------------------------------|--------|
| Employee login             | Successful login within acceptable response time      | Passed |
| Submit onboarding request  | Request submitted and workflow initiated successfully | Passed |
| Submit offboarding request | Request processed without delay                       | Passed |
| Approval workflow          | Approvals routed correctly to HR, Manager, and IT     | Passed |
| Account provisioning       | User account created successfully                     | Passed |
| Account deactivation       | User account disabled after approval                  | Passed |
| Email notifications        | Notifications delivered successfully                  | Passed |
| Dashboard loading          | Dashboard displayed without noticeable delay          | Passed |
| Report generation          | Reports generated accurately and efficiently          | Passed |

## Performance Metrics

| Parameter             | Result              |
|-----------------------|---------------------|
| Average Response Time | Less than 3 seconds |
| Workflow Execution    | Successful          |
| Database Performance  | Stable              |
| System Availability   | High                |
| Error Rate            | Minimal             |

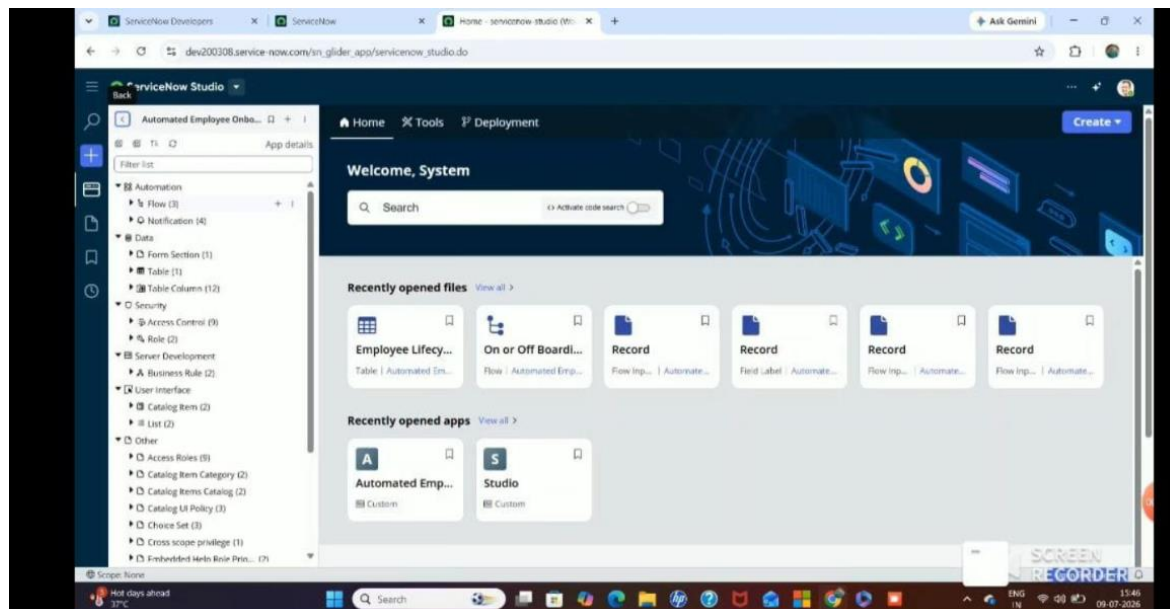
|                 |                       |
|-----------------|-----------------------|
| User Experience | Smooth and Responsive |
|-----------------|-----------------------|

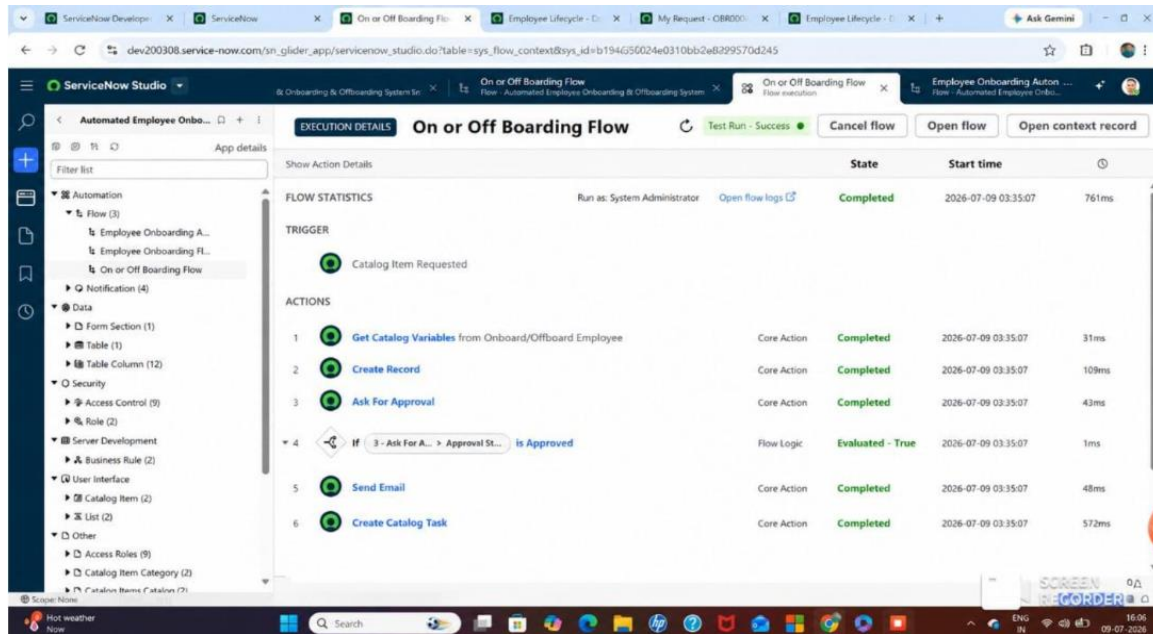
## Testing Outcome

The performance testing results indicate that the **Automated Employee Onboarding & Offboarding System** meets the expected performance requirements. All major workflows, including onboarding, offboarding, approval processing, account management, and notifications, executed successfully. The application demonstrated reliable performance, quick response times, and stable operation under typical usage conditions, making it suitable for deployment in an organizational environment.

# 7. RESULTS

## 7.1 Output Screenshots





## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Automates employee onboarding and offboarding processes, reducing manual effort.
- Saves time by streamlining approvals and task assignments.
- Minimizes human errors through standardized workflows.
- Improves productivity for HR, IT, and management teams.
- Enhances security by automatically granting and revoking system access.
- Maintains accurate employee records in a centralized database.
- Provides real-time tracking of onboarding and offboarding progress.
- Sends automated email notifications and reminders.
- Improves compliance through audit trails and activity logs.
- Enhances the overall employee experience with faster and more efficient processes.
- Supports scalability, making it suitable for organizations of different sizes.
- Generates reports and dashboards for better decision-making.

## Disadvantages

- Initial setup and configuration require time and technical expertise.
- Implementation costs may be high for small organizations.
- Requires user training for HR staff and employees.
- Depends on stable internet connectivity and system availability.
- Integration with third-party applications may require additional configuration.
- Workflow customization can increase development and maintenance effort.
- System updates and maintenance are necessary to ensure optimal performance.
- Temporary system downtime may affect onboarding or offboarding activities.
- Incorrect configuration of workflows may lead to processing delays.
- Data migration from legacy systems can be complex and time-consuming.

Overall, despite a few implementation and maintenance challenges, the advantages of the **Automated Employee Onboarding & Offboarding System** significantly outweigh its disadvantages by improving efficiency, security, compliance, and the employee experience.

## 9.CONCLUSION

The **Automated Employee Onboarding & Offboarding System** successfully streamlines and automates the employee lifecycle by replacing manual processes with efficient digital workflows. Developed using the ServiceNow platform, the system simplifies onboarding and offboarding activities such as employee registration, approval management, account provisioning, asset tracking, document management, and access control.

By automating routine tasks, the system reduces processing time, minimizes human errors, enhances security, and ensures compliance with organizational policies. Real-time notifications, dashboards, and reporting features improve transparency and enable HR, IT, and management teams to monitor progress effectively.

Overall, the project demonstrates how workflow automation can improve operational efficiency, enhance the employee experience, and strengthen organizational security. The **Automated Employee Onboarding & Offboarding System** provides a scalable, reliable, and user-friendly solution that can be further enhanced with advanced technologies such

as Artificial Intelligence (AI), predictive analytics, chatbot assistance, and integration with external HR systems to support future organizational needs.

## 10.FUTURE SCOPE

The **Automated Employee Onboarding & Offboarding System** can be further enhanced by incorporating advanced technologies and additional features to improve efficiency, user experience, and organizational productivity. As business requirements evolve, the system can be expanded to support more intelligent and integrated HR operations.

### Future Enhancements

- Integrate **Artificial Intelligence (AI)** to automate employee support and provide intelligent recommendations.
- Add **chatbot integration** to answer employee queries and guide users through onboarding and offboarding processes.
- Implement **predictive analytics** to identify workflow bottlenecks and improve process efficiency.
- Integrate with external **HRMS, Payroll, ERP, and Active Directory** systems for seamless data synchronization.
- Develop a **mobile application** to allow employees and managers to access services anytime and anywhere.
- Introduce **e-signature support** for faster approval of documents and agreements.
- Enhance reporting with **interactive dashboards** and real-time analytics for better decision-making.
- Implement **multi-factor authentication (MFA)** and advanced security features to strengthen data protection.
- Support **multi-language interfaces** to accommodate a diverse workforce.
- Enable **cloud-based scalability** to support organizations with multiple branches and a growing number of employees.
- Add automated compliance monitoring and audit reporting to meet regulatory requirements.
- Incorporate employee feedback and satisfaction surveys to continuously improve onboarding and offboarding experiences.

